

Evaluating Pretrained Architectures for Agricultural Pest Classification Across Diverse Datasets

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Abstract. Pests can lead to significant environmental and economic loss, making accurate identification a crucial first step in effective agricultural management strategies. Modern deep learning models for image classification can assist in this task and improve accuracy. This work evaluates the generalization capabilities of two broad, pretrained (on ImageNet) architectures: a convolutional neural network (ResNet-50) and a vision transformer (ViT-B/16), across diverse pest datasets. Both models are tested on two datasets, *Dangerous Farm Insects* and *Pest Dataset V2* under zero-shot and fine-tuning settings for comparison. While overall performance was similar between the two models, the necessity for domain-specific training is apparent, and their differences arose through training capabilities and limitations.

Keywords: Deep learning, classification, convolutional neural network, vision transformer, zero-shot, fine-tuning, transfer learning

1 Introduction

Pests are a consistent problem that farmers, gardeners, and other agricultural professionals face. Damage from pests can not only negatively impact crop quality but can also lead to total crop loss. To minimize crop pest infestation and maximize pesticide efficiency and effectiveness, professionals need a method of identifying what the pest is, when to spray pesticides, what pesticides to utilize, and how much. The goal is to minimize the cost and use of pesticides while maintaining a healthy and plentiful agricultural area via pest control.

In the past, farmers have relied on human identification, but that can be tedious and unreliable. With advancements in technology, machine learning and deep learning models have been developed to assist with this task. Although existing pest classification models are able to obtain high accuracy, many are trained on specific datasets and struggle to generalize to new environments. These models can also be sensitive to real-world conditions, like background noise, and struggle to perform well when presented with image variability. Additionally, there are gaps in actual model architectures. There are not a lot of existing models that implement attention or transformers.

This project aims to provide greater understanding about existing methods and models to successfully identify pests and therefore provide helpful insight

on what necessary actions to take in order to combat and prevent damage from infestations. Specifically, we will be analyzing the performance of two existing model designs (convolutional neural networks (CNNs) vs Visual Transformers (ViTs)) on multiple new datasets by using evaluation metrics such as classification accuracy, precision, recall, and F1-score for each pest class. We are providing a comparative analysis that will help reveal how well different architectures perform on real-world, noisy pest image datasets. In other words, which architecture generalizes better?

2 Related Work

Machine Learning to Deep Learning for Insect Classification. Before the deep learning era, pest identification relied on traditional machine learning and computer vision techniques using handcrafted features [1]. Deep learning models, particularly CNNs and advanced variants, now dominate this task due to improved performance and feature learning capabilities [2]. Since a decent amount of work already exists around creating new and improving model architectures and performance, rather than proposing new architectures, we aim to compare existing models and focus on how well they generalize across datasets and real-world conditions.

CNNs for Pest Classification. CNN architectures, particularly when combined with transfer learning, have historically performed well on pest classification tasks [3, 4]. Models such as ResNet and EfficientNet achieve high accuracy on domain-specific datasets focused on a single pest category or controlled setting [5]. However, these works do not systematically evaluate how well models generalize to new datasets comprising images with real-world backgrounds that introduce noise and distribution shift. In contrast, our work addresses this gap through cross-dataset evaluation.

Vision Transformers for Pest Classification. Transformer-based models have been increasingly applied to pest classification, demonstrating improvements in capturing global context that CNNs may miss [6–9]. These works primarily focus on designing new or improved architectures to boost performance on a fixed benchmark. In contrast, we do not propose new architectures; instead, we analyze whether existing ViT models can generalize across pest datasets without architectural modification, and compare their performance directly against CNN baselines.

Generalization and Evaluation of Pretrained Models. Large-scale datasets such as ImageNet have played a critical role in advancing deep learning models for image classification and enabling the development of powerful pretrained models that are widely used across domains [10]. However, the degree to which these models generalize to specialized tasks remains an open question—even within the ImageNet distribution itself [11]. Our work is different from these works because

we directly evaluate generalization in the context of agricultural pest classification by testing pretrained models across different pest datasets, of varying sizes and class distributions, rather than a single benchmark dataset. We also want to further investigate how training these models on smaller, domain-specific pest data improves classification performance.

3 Methodology

3.1 Models

The two existing model designs evaluated in this project are a CNN and a ViT model architecture. CNN-based models have historically served as a strong baseline for pest classification tasks and are among the most well-studied architectures in this domain [6]. More recently, ViT models have emerged as a more competitive alternative when pretrained on large-scale datasets. There is also evidence that these models were experimented with as a method to improve results while simultaneously requiring fewer computational resources [8].

For the CNN baseline, we are using a pre-trained model called ResNet-50, which has 50 convolutional layers where weights were initialized from ImageNet [10]. This model is widely used in prior pest classification studies, allowing us to compare directly to prior work. The model is sourced from PyTorch and TensorFlow/Keras and is adapted for our task by replacing the final classification layer with a fully connected layer to match the number of output classes. First, no additional training will be performed, and the pretrained weights are used directly for evaluation. The model is then adapted with fine-tuning layers to train on a subset of the pest datasets.

For the transformer-based approach, we will use the ViT-B/16 model, also pre-trained on ImageNet, which follows the standard baseline from Dosovitskiy et al. [8] with 16×16 pixel patches and is available on HuggingFace and PyTorch. Similar to the CNN, the final classification head is changed to match the number of classes. Similarly, the model will first be evaluated with no training then will be fine-tuned on the target datasets.

3.2 Datasets

To ensure a meaningful comparison, we are conducting experiments on two new datasets. The first dataset, *Dangerous Farm Insects* (Kaggle), contains 1,591 relatively clean images with 15 classes listed in Table 1 (just over about 100 images per class). This dataset is split into a train, validation, and test set (70-15-15) for Experiment 2 described below.

Africanized Honey Bee	Aphids	Armyworms
Brown Marmorated Stink Bugs	Cabbage Loopers	Citrus Canker
Colorado Potato Beetles	Corn borers	Corn Earworms
Fall Armyworms	Fruit Flies	Spider Mites
Thrips	Tomato Hornworms	Western Corn Rootworms

Table 1: Pest Classes in Dangerous Farm Insects

The second, larger dataset, *Pest Dataset V2* (Kaggle), consists of 34,200 images with 19 classes (see Figure 1) already split into train (1,500 per class), test (150 per class), and validation (150 per class) datasets. The images were sourced from multiple public datasets, allowing evaluation with more diverse and noisy images. These datasets allow us to compare model performance for controlled and realistic conditions, and fine-tuning.

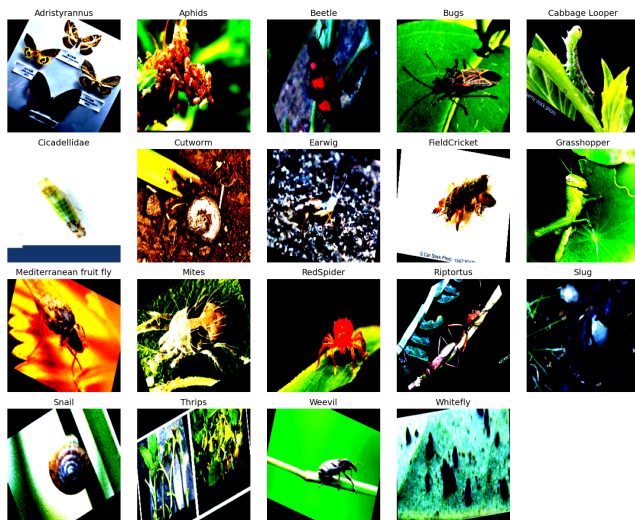


Fig. 1: Example images from each of 19 classes in Pest Dataset V2.

3.3 Data Preprocessing

All images are resized such that the shorter size is 256 pixels and then center-cropped to 224×224 pixels to match the input requirements of both ResNet-50 and ViT-B/16. The images are then converted to tensors and normalized using ImageNet mean and standard deviation. No augmentation is applied at first since the models are evaluated without training. Then later, augmentation is performed to conduct the fine-tuning and the transfer learning process of the

pest-specific datasets. Identical preprocessing is applied across both datasets to ensure a reasonable comparison.

4 Experiments

4.1 Experiment 1: Generalization Across Datasets

Main Purpose. To evaluate how well pretrained image classification models generalize across new pest classification datasets that differ in scale, class distribution, and background noise, we will load and test ResNet-50 and ViT-B/16 on two datasets: Dangerous Farm Insects and Pest Dataset V2 using the same pre-processing described above. Results will be compared across datasets and model types (CNN vs ViT) to assess versatility and determine which architecture performs better for this task.

Evaluation Metrics. Classification accuracy will be used for overall performance comparisons and macro F1-score is computed to account for class imbalance in datasets. Precision and recall will be computed per class and averaged for the F1-score. We will also include confusion matrices to identify potential common misclassifications between visually similar pest species, providing additional contextual insight.

Dangerous Farm Insects Dataset. The ViT model barely outperformed the ResNet backbone shown by the confusion matrices in Figure 2 where ViT has a stronger diagonal indicating stronger correct classification. In general, both models performed as expected, failing to generalize. ResNet-50 obtained an accuracy of 0.10 whereas ViT-16/B obtained an accuracy of 0.04. The difference between classification accuracy is so small, it is inconclusive which model is better at generalizing without any task-specific fine-tuning or transfer learning. While both models demonstrated some understanding of general image features, the accuracy remained significantly low due to both models pre-trained on images that help the model capture general features, lacking the ability to generalize to pest-specific data.

The confusion matrix for zero shot performance for ResNet-50 shows certain classes are consistently misclassified as others. This could likely be due to the general purpose weights from ImageNet, that lack the ability to distinguish other classes further.

The visual transformer matrix appears to be slightly more robust to variations in the zero shot environment. This is likely due to architectural differences between the CNN and visual transformer. The attention mechanism in the transformer allows the model to pick up broader details and context that a CNN cannot.

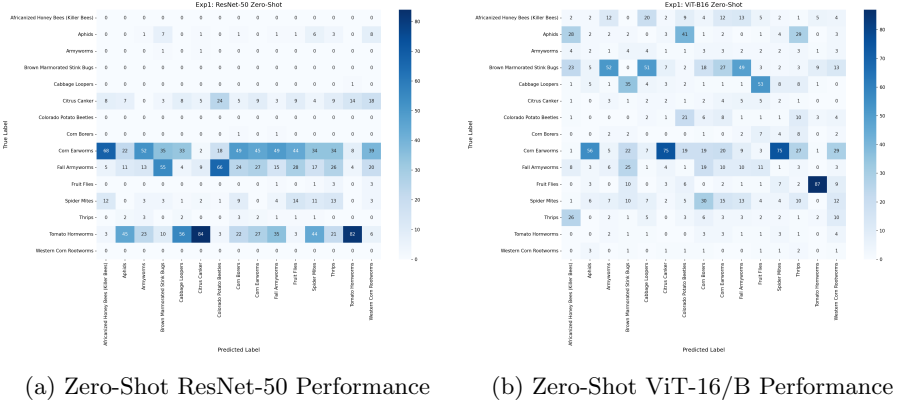


Fig. 2: Experiment 1 results for the Dangerous Farm Insects Dataset

Pest Datasets V2. Both models, ResNet-50 and ViT-B/16, evaluated on the larger test dataset of 2,800 images with 19 pest classes, performed just as poorly, achieving accuracies of 0.048 and 0.059, respectively. While ViT-B/16 slightly outperformed ResNet-50 based on accuracy and F1 scores, the difference was minimal, only about 1%. This slight improvement of ViT-B/16 on the larger dataset, compared to its performance on the smaller dataset, may align with the idea that transformer-based models benefit more from larger amounts of data. Though, as both models were effectively making random predictions, limited comparison can be made.

Similarly to the first dataset, looking at the confusion matrices (Figure 3), however, reveals differences in how each model makes predictions in the zero-shot setting. ResNet-50 shows a strong bias towards a few classes, predominately predicting *Riptortus* for many images (including Bugs, Earwig, and Adristyrannus), indicated by the prominent line in the matrix, and frequently misclassified *snails* as *Weevil*, indicated by the off-diagonal dark blue square. In contrast, ViT-B/16 produced more scattered, though still largely incorrect, predictions, with a common confusion between *Weevil* and *Cicadellidae*. Again, this suggests that the Vision Transformer is more robust in its "guessing" than the CNN.

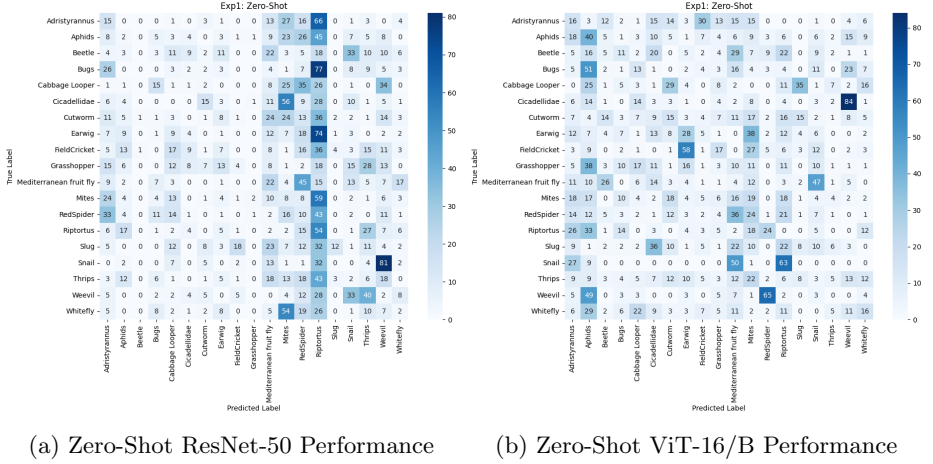


Fig. 3: Experiment 1 results for the Pest Dataset V2

Notably, *Weevil* was heavily predicted by both models, which may be explained by it being among the ImageNet classes used during pretraining. We see both models had to rely on familiar features from pretraining, rather than being able to effectively distinguish pest-specific categories in this zero-shot setting.

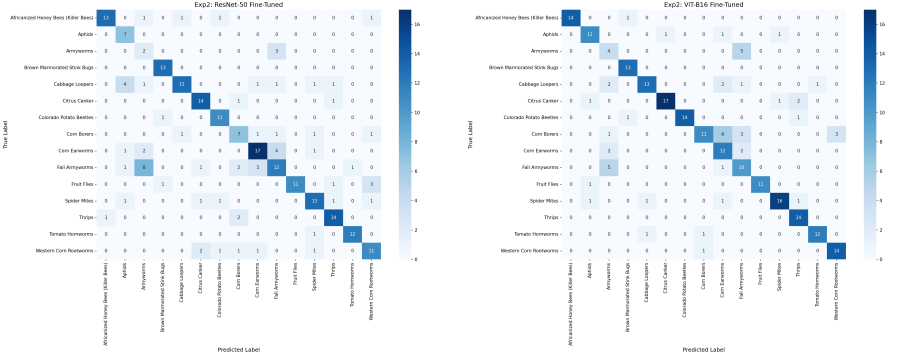
4.2 Experiment 2: Fine-Tuning on Target Datasets

Main Purpose. The goal of this experiment is to explore the effect of fine-tuning pretrained models on domain-specific pest datasets and compare performance with zero-shot evaluation. Both ResNet-50 and ViT-B/16 will be fine-tuned on a subset of each dataset and evaluated on the remaining data, with the final classification layer adapted to match the dataset classes.

Evaluation Metrics. We will use accuracy and macro F1-score for consistency and comparison with Experiment 1. Confusion matrices are included to analyze common misclassifications among pest species. We will compare performance between pretrained and fine-tuned models to quantify the impact of domain-specific training. We will also report training accuracy to assess differences in learning behavior between CNN and ViT models for pest classification tasks.

Dangerous Farm Insects Dataset. Recall that this dataset is divided into a train, validation, and test set with a split of 70/15/15. Both the CNN and ViT model are trained on the train set in order to perform fine-tuning of the domain-specific dataset. The validation set is then used to evaluate if the model is generalizing on unseen data and allows full fine-tuning of the network's weights with the Adam optimizer to minimize loss as well as fine-tuning the learning rate

with a scheduler. It is important to note that early stopping was implemented to prevent overfitting and memorization of training data.



(a) Fine-Tuned ResNet-50 Performance (b) Fine-Tuned ViT-16/B Performance

Fig. 4: Experiment 2 results for the Dangerous Farm Insects Dataset

There is a significant shift in both models' performance as shown in Figure 4. It is clear that the confusion matrices show a cleaner diagonal compared to Experiment 1. This is strong evidence that fine-tuning allows the models to map high level features and details learned during training to the specific class labels.

More specifically, the ResNet-50 model shows significant improvement after fine-tuning. After specializing the model to this pest dataset, this CNN model architecture achieved an accuracy of 0.7143 and a macro F1 score of 0.7121 with early stopping occurring at epoch 8. Thus, it is proven that it can be highly effective after fine-tuning to the specific pest dataset.

Similarly, the fine-tuned ViT-B/16 model performed exceptionally better compared to zero shot performance. This fine-tuned model achieved a classification accuracy of 0.7815 and a macro F1 score of 0.7857 with early stopping occurring at epoch 10. Relative to the fine-tuned CNN, the visual transformer model slightly outperformed the CNN as seen by the higher accuracy. However, limitations to exceeding even greater robust accuracy include not enough training and overfitting to the small dataset. A more in-depth analysis on training is below to understand training dynamics.

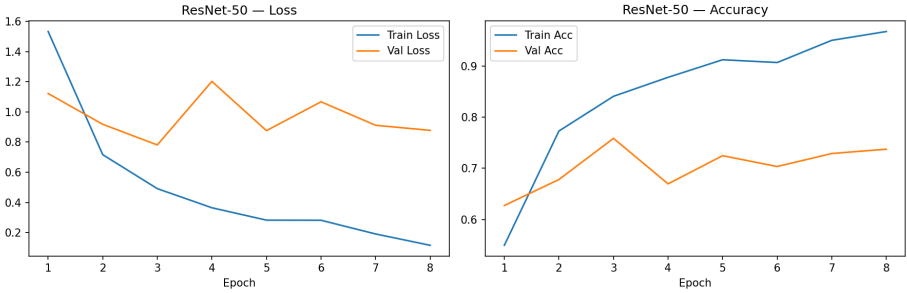


Fig. 5: Train and validation loss (left) and accuracy (right) curves for ResNet-50

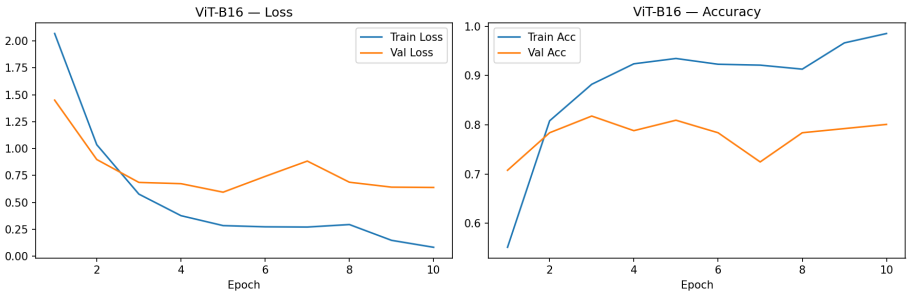


Fig. 6: Train and validation loss (left) and accuracy (right) curves for ViT-B/16

As seen in Figure 5, the training curves for ResNet-50 show steady decline in loss. In contrast, the ViT-B/16 curves in Figure 6 show faster convergence in training accuracy since the curve is steeper and increases at a faster rate. Additionally, Both models show signs of overfitting. Toward the end of the epochs, the training and validation loss for both models (left plots in Figures 5 and 6) show a little divergence. This gap between loss curves is relatively controlled which means data augmentation before training and the scheduler for the learning rate were somewhat effective. To improve accuracy of these models, more training data is required.

Pest Dataset V2. The same training methods from the first dataset, validation, Adam optimizer, learning rate scheduler, and early stopping, were implemented. After fine-tuning on *Pest Dataset V2*, with over 28,000 training images, both models showed a significant improvement in performance compared to the zero-shot setting in Experiment 1 and achieved higher accuracy than on the smaller dataset. Both ResNet-50 and ViT-B/16 reached approximately 84% accuracy and macro F1 scores with a learning rate of 0.001. The confusion matrices in Figure 7 again exhibit strong diagonal patterns, indicating consistent and accurate predictions, with no major recurring misclassifications. However,

interestingly, there was a similar minor confusion between the grasshopper and beetle in both models.

Performance was similar between the two architectures, with no significant difference in accuracy or F1-scores, but differences emerged in training behavior. As seen in Figures 8 and 9, ViT-B/16 converged in fewer epochs, while ResNet-50 required more epochs to reach comparable performance. ResNet-50 started at less than 75% training accuracy and required 6 epochs to reach about 90% accuracy. The best model was saved after 4 epochs with little to no improvement in validation accuracy (and increased validation loss) for 3 epochs resulting in early stopping after epoch 7. While ViT-B/16 started with a training accuracy of 80% and reached above 90% after only 3 epochs, and early stopped after 6 epochs.

Although the vision transformer model resulted in faster training, requiring less epochs, it had higher computational cost per epoch, resulting in longer overall runtime (more than triple the time for ResNet-50). Being a larger dataset, training for both models resulted in significantly longer run-times but faster and more stable training than with the *Dangerous Farm Insects Dataset*.

Note, due to GPU runtime constraints, transfer learning was employed, where all layers but the last layer were frozen and only the final classification layer was trained with the larger dataset, instead of fully fine-tuning the entire network like with the first dataset. Even with this partial fine-tuning, both models achieved better performance with accuracy increased by about 10% with the larger dataset. Although full fine-tuning on *Pest Dataset V2* may have given the best results overall (higher than 85% accuracy), this highlights the effectiveness and practical advantages of transfer learning for real-world applications where computational resources may be limited.

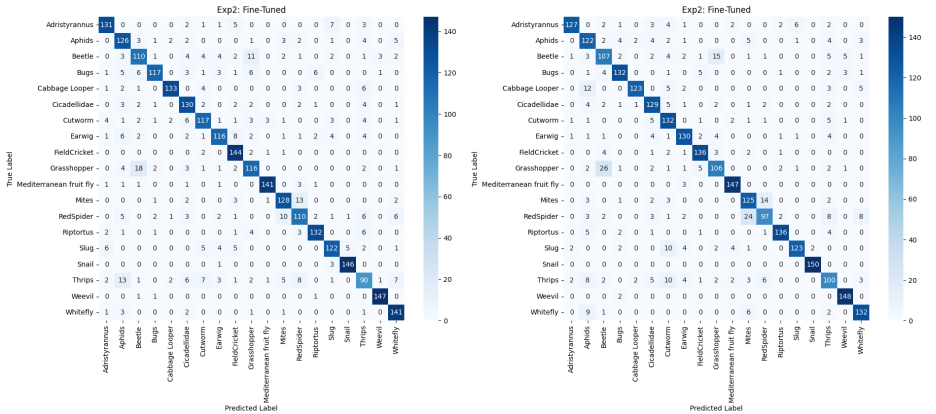


Fig. 7: Experiment 2 results for the Pest Dataset V2

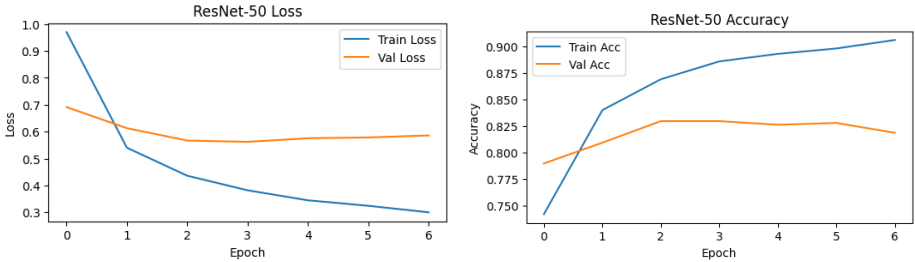


Fig. 8: Train and validation loss (left) and accuracy (right) curves for ResNet-50

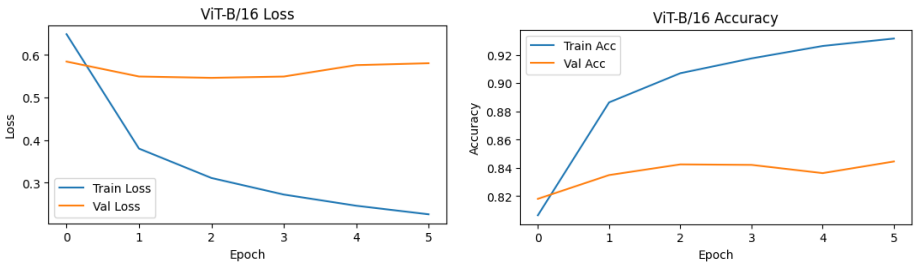


Fig. 9: Train and validation loss (left) and accuracy (right) curves for ViT-B/16

4.3 Experiment and Dataset Comparison

Table 2 displays a summary of the results for both experiments on the Dangerous Farms Insects Dataset.

Model	Method	Accuracy	Macro F1-Score
ResNet-50	Zero-shot	0.1007	0.0446
ViT	Zero-shot	0.0443	0.0419
ResNet-50	Finetuned	0.7143	0.7121
ViT	Finetuned	0.7815	0.7857

Table 2: Performance Comparison of ResNet-50 and ViT models (Zero-shot vs. Finetuned) on the Dangerous Farm Insects Dataset

Table 3 displays a summary of the results for both experiments on the Pest Dataset V2.

Model	Method	Accuracy	Macro F1-Score
ResNet-50	Zero-shot	0.0477	0.0348
ViT	Zero-shot	0.0589	0.0546
ResNet-50	Finetuned	0.8411	0.8403
ViT	Finetuned	0.8428	0.8428

Table 3: Performance Comparison of ResNet-50 and ViT models (Zero-shot vs. Finetuned) on the Pest Dataset V2

Fine-tuning resulted in a substantial improvement in performance across both datasets, increasing accuracy from approximately 4–10% in the zero-shot setting to 70–84% after training. This confirms that pretrained models alone do not generalize well to domain-specific pest datasets and require adaptation through fine-tuning. The poor zero-shot performance can be attributed to the mismatch between ImageNet pretraining and pest-specific classes, as the models are not trained to recognize most pest categories directly and rely only on general feature representations.

On the larger *Pest Dataset V2*, both ResNet-50 and ViT-B/16 achieved their best performance, with nearly identical accuracy (84%), suggesting that dataset size plays a critical role in model effectiveness.

While overall performance was similar, important differences emerged in training dynamics. ViT-B/16 required fewer epochs to reach high accuracy, indicating faster learning, whereas ResNet-50 had shorter per-epoch runtime and was more computationally efficient. This highlights a trade-off between convergence speed and computational cost, which may influence model selection depending on deployment constraints.

Additionally, differences between accuracy and macro F1-score suggest that ViT-B/16 handles class imbalance more effectively in the zero-shot setting, whereas ResNet-50 is more sensitive to imbalance without training but performs comparably after fine-tuning.

Experimental Limitations and Further Work. While our experiments provide useful insights, several questions remain unanswered. First, although these pretrained models were found to not generalize well to pest classification in a true zero-shot setting, this was expected largely due to the mismatch between ImageNet labels, what the models were pretrained on, and the new pest-specific target labels. There was little to no overlap in labels, highlighting the importance of domain-specific training, but making it inconclusive on true generalization capabilities. Additional experiments, such as training on one pest dataset and testing on another, would further evaluate cross-dataset generalization in a domain-specific setting. Further analysis on what features these pretrained models failed to capture and why certain misclassifications occurred could also bring insight into these short comings. Future work could also explore alternative zero-

shot approaches or models for pest-classification that better align pest-specific image features with target labels.

Additionally, since both architectures achieved similar performance after fine-tuning, it remains unclear which model type generalizes better overall. With more compute power, experiments on larger, more diverse, such as IP102, and noisier datasets would help clarify differences in robustness and improve real-world applicability. Investigating how performance scales with dataset size and data quality would also provide insight into the data efficiency of CNNs versus Vision Transformers.

Another open question is how much training is truly necessary. There is a gap between zero-shot inference, transfer learning, and full fine-tuning that was not fully explored in this work. Future studies could examine few-shot learning and partial fine-tuning strategies to better understand this trade-off.

5 Conclusions

Pretrained models alone do not generalize well to real-world pest datasets, as shown by the poor pure zero-shot performance. However, with some level of fine-tuning, both ResNet-50 and ViT-B/16 achieved strong results, demonstrating the importance of domain-specific training. While performance was similar across models, the Vision Transformer showed slightly better generalization overall.

For specific tasks like pest classification, we found large pretrained models can be an effective choice when fine-tuned properly. They can save time and effort without requiring extensive expertise compared to building models from scratch, and can ultimately lead to effective pest management decision-making. However, misclassifications can lead to the misuse of pesticides or incorrect preventative measures, potentially causing significant crop damage, financial loss, and environmental harm. This directly highlights the need for accountability and transparency in their use. Understanding the limitations of these models is crucial and users should not rely on them solely for automated decision-making. Rather, users should apply these models as a tool alongside human supervision and expertise to help guide decisions. Additionally, issues with implementation and accessibility arise as technology often requires resources, money, and technical knowledge that many farmers may lack.

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